

NGR Lowline Auto 1.0 – Final Design Summary

Architecture:

Fully onboard ESP32 locomotives using ESP-NOW peer-to-peer communication. No trackside electronics.

Base Code:

GoldCore v9 renamed to LowLine Auto, with Highline logic removed.

Modes:

Manual, Auto, E_STOP.

Startup:

Power up stopped → manual setup → Auto enabled → first valid magnet sets position.

Hall Sensor:

US1881 bipolar, GPIO 33, open collector with pull-up.

Magnet Code System:

3-magnet polarity groups using full 3-bit encoding (NNN, NSN, NNS, NSS, SSS, SNS, SSN, SNN).

Route Order:

Stage Left → Stage Right → Cairo → Grillers → Corner → Camelia → Buddha.

Communication:

ESP-NOW every 500 ms. STOPPED sends 3-packet burst then continues.

Status:

GO = moving. STOPPED = PWM = 0 after ramp-down.

Speed:

Cruise 120, Approach 80, Compression 50. No brake pulse.

Interaction:

Peer STOPPED in next block → delay 3–5 sec → slow to 80. Deep compression → 50.

Station Stops:

Ramp down → stop → dwell 5–15 sec → ramp up.

Failsafe:

If STOPPED exceeds dwell + margin → following train stops.

Philosophy:

Distributed, reactive, realistic, expandable.